

LoRA Uni-LoRA ProLoSA Oracle Uni-LoRA

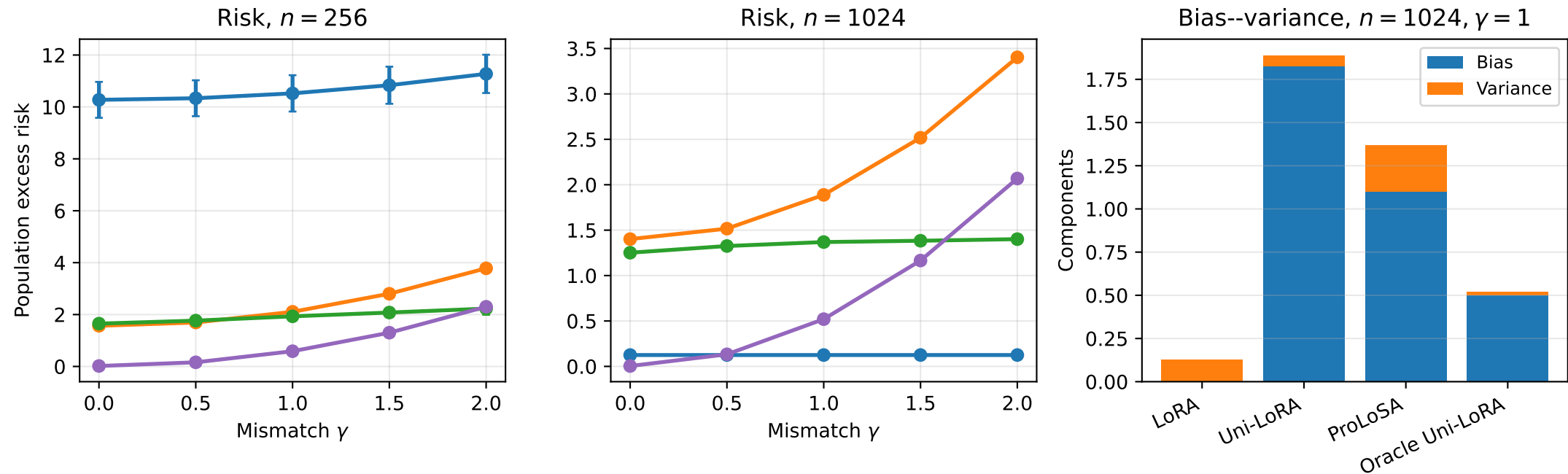


Figure S2: Hidden-P non-oracle projection with controlled mismatch. The teacher uses P_T to generate the update, while Uni-LoRA and ProLoSA only see P_M , here a 15-degree rotation of P_T (mean squared canonical overlap 0.933). ProLoSA remains below Uni-LoRA for larger gamma at $n=256$ and $n=1024$; oracle Uni-LoRA with P_T is shown only as an upper bound.